Directed Numbers and Zero-Point Operators: An Algebraic Formalism for Topological Information Conservation

NOWN Research Collective
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Abstract

We introduce a novel algebraic formalism—**directed numbers** $a^{\uparrow}, a^{\downarrow}, a^{0}$—designed to track both amplitude and topological parity in information systems. This algebra is motivated by geometric features of non-orientable manifolds (Möbius strips, Klein bottles), where "zero" is interpreted not as absence but as a compressed singularity. We define compression (Ω) and expansion (Ω^{-1}) operators and show that for multi-component systems, information conservation constrains admissible transformations; in particular, they can be represented by doubly stochastic matrices, with the Sinkhorn–Knopp algorithm providing a natural projection onto the Birkhoff polytope.

This paper presents a refined axiomatic foundation for directed numbers, incorporating:

- The distinction between absolute zero and directed zeros with memory ($0^{\uparrow}, 0^{\downarrow}, 0_{\text{abs}}$)
- Complete multiplication tables including probabilistic outcomes for absolute-zero products
- Non-associativity as a feature encoding path-dependence in non-orientable spaces
- Thread calculus for parallel and nested contexts
- Temporal extensions for modeling information flow across time
- **New in v0.8:** Modular parity framework allowing extension to additional states (e.g., gravitational chirality); retrocausal interpretation as parallel thread calculus; temporal consistency conditions linking past and future zero-point products; connection to complex numbers and Hilbert spaces; implications for AI and self-referential computation.

The formalism is presented as a self-contained mathematical tool, with potential applications across multiple domains (quantum mechanics, cosmology, artificial intelligence, protein folding) developed in dedicated companion papers.

Note on Status

This document describes work in progress (v0.8). The directed number formalism is mathematically specified, but its physical interpretation and empirical validation remain open areas of investigation. We invite collaboration, critique, and refinement from the broader scientific community.

1 Introduction

The concept of zero in conventional mathematics denotes absence—a void where information is destroyed (multiplication by zero) or operations become undefined (division by zero). However, in non-orientable geometries such as Möbius strips and Klein bottles, the "twist" creates a singularity where information is not destroyed but *compressed*, retaining its content while becoming inaccessible to the local orientation of the manifold.

This geometric observation motivates a new algebraic structure: **directed numbers**, which explicitly track the parity (orientation) of a state alongside its amplitude. In this formalism, multiplication by zero becomes a compression operation that moves information into the singularity, and division by zero becomes an expansion that retrieves it, potentially with inverted parity.

The algebra is self-contained and mathematically rigorous. Its interpretation in terms of physical systems—from quantum mechanics to protein folding to consciousness—is developed in separate publications (see companion papers on Information Substrate Theory [1]). Here, we focus on the formalism itself, presenting it as a tool that researchers across disciplines can adapt to their own domains.

In this updated version (v0.8), we emphasize the **modularity** of the formalism: the core parity set $\{\uparrow, \downarrow, 0\}$ can be extended as needed for specific applications (e.g., gravitational states $\circlearrowleft, \circlearrowright$ in the IST context). We also introduce a **retrocausal interpretation** of the probabilistic axiom, framing it as a parallel thread calculus where future boundary conditions determine present outcomes—a rigorous mathematical feature with potential applications in distributed computing and self-referential systems.

2 Axiomatic Foundation

2.1 Elements and Parity

Axiom 2.1 (The Set)**.** *The set of directed numbers $\mathbb{D}$ consists of elements of the form a^p, where:*

- *$a \in \mathbb{R}$ (or $\mathbb{C}$) is the **amplitude***
- *$p \in \{\uparrow, \downarrow, 0\}$ is the **parity***

We denote $a^\uparrow$ as up-manifest, $a^\downarrow$ as down-manifest, and a^0 as compressed (unmanifest) state.

Axiom 2.2 (Modular Parity Extension)**.** *The parity set $\{\uparrow, \downarrow, 0\}$ can be extended to include additional states as needed for specific applications. For example, in the Information Substrate Theory framework, gravitational chirality states $a^\circlearrowleft$ and $a^\circlearrowright$ are introduced to model dark matter as gravitational solitons. Such extensions must preserve the core algebraic properties (information conservation, non-associativity) and define consistent multiplication rules for the new states.*

Axiom 2.3 (Zero as a Special Amplitude)**.** *The amplitude 0 may appear with any parity:*

- *$0^\uparrow$ and $0^\downarrow$ are **directed zeros**—they carry memory of having been compressed from a manifest state of known parity.*
- *0^0 (or 0_{abs}) is **absolute zero**—the pristine zero-point gate with no history.*

Directed zeros are distinct from absolute zero; they encode topological memory essential for information conservation.

2.2 Addition

Axiom 2.4 (Same-Parity Addition)**.** *For any parity $p \in \{\uparrow, \downarrow, 0\}$, addition is defined componentwise:*

$$a^p + b^p = (a + b)^p.$$

Axiom 2.5 (Mixed-Parity Addition). *Addition of elements with different parities is not directly defined. Such sums represent* ***superpositions*** *that require additional context (e.g., a choice of basis or a mediating operation). They may be interpreted as probabilistic mixtures pending future refinement.*

2.3 Multiplication

Axiom 2.6 (Multiplication by Real Scalars). *For any real $\lambda \in \mathbb{R}$ and any directed number a^p:*

$$\lambda \cdot a^p = (\lambda a)^p.$$

Axiom 2.7 (Product of Manifest Numbers). *For manifest numbers ($p, q \in \{\uparrow, \downarrow\}$),*

$$a^p \cdot b^q = \begin{cases} (ab)^p & \text{if } p = q, \\ (ab)^0 & \text{if } p \neq q. \end{cases} \tag{1}$$

Axiom 2.8 (Product Involving Compressed Numbers). *For any directed number a^p and any compressed number b^0 (including directed zeros and absolute zero):*

$$a^p \cdot b^0 = (ab)^0.$$

Similarly, $b^0 \cdot a^p = (ab)^0$.

Axiom 2.9 (Product of Two Compressed Numbers). *This product depends on the specific type of compressed numbers:*

- ***Two directed zeros of the same memory:***

  $$(0^\uparrow) \cdot (0^\uparrow) = 1^\uparrow, \quad (0^\downarrow) \cdot (0^\downarrow) = (-1)^\downarrow.$$

  (More generally, if directed zeros are understood as compressions of manifest amplitudes a^p and b^p, then the manifest output has amplitude ab, with a sign determined by the memory/parity.)

- ***Two directed zeros of opposite memory:***

  $$(0^\uparrow) \cdot (0^\downarrow) = 0^0 \quad (\textit{absolute zero}).$$

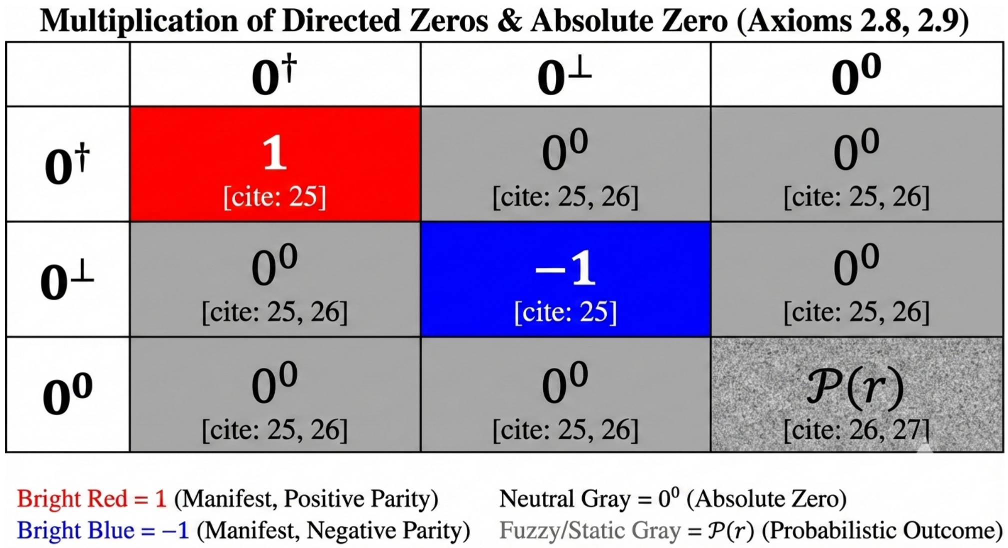
- ***Any product involving absolute zero:***

  $$0^0 \cdot 0^0 = \textit{? (probabilistic; see Axiom 2.10).}$$

  $$0^0 \cdot 0^\uparrow = 0^0 \cdot 0^\downarrow = 0^0.$$

Axiom 2.10 (Probabilistic Outcome for Absolute-Zero Products). *The product $0^0 \cdot 0^0$ is not deterministic. It yields a real number $r \in [-1, 1]$ drawn from a probability distribution $\mathcal{P}(r)$ that is symmetric about 0 and has variance related to the topology. The exact form of $\mathcal{P}$ is an open question; candidate distributions include the uniform distribution, the normal distribution, or a distribution derived from the golden ratio fixed point.*

In v0.8, we interpret this randomness as ***epistemic*** *rather than ontological: the outcome is determined by future boundary conditions that are inaccessible from the present reference frame (see Axiom 2.18). In computational terms, this corresponds to a parallel thread that computes a needed value and inserts it retrocausally into the current equation—a mathematically rigorous feature with applications in distributed and self-referential computing.*



Multiplication of Directed Zeros & Absolute Zero (Axioms 2.8, 2.9)

	$0^\dagger$	$0^\perp$	0^0
$0^\dagger$	1 [cite: 25]	0^0 [cite: 25, 26]	0^0 [cite: 25, 26]
$0^\perp$	0^0 [cite: 25, 26]	-1 [cite: 25]	0^0 [cite: 25, 26]
0^0	0^0 [cite: 25, 26]	0^0 [cite: 25, 26]	$\mathcal{P}(r)$ [cite: 26, 27]

Bright Red = 1 (Manifest, Positive Parity)
Neutral Gray = 0^0 (Absolute Zero)
Bright Blue = -1 (Manifest, Negative Parity)
Fuzzy/Static Gray = $\mathcal{P}(r)$ (Probabilistic Outcome)

Figure 1: Multiplication behavior for directed zeros and absolute zero, illustrating the probabilistic outcome $\mathcal{P}(r)$ defined in Axiom 2.10.

2.4 Zero-Point Operators

Axiom 2.11 (Compression Operator Ω). *For any manifest number a^p ($p \in \{\uparrow, \downarrow\}$),*

$$\Omega(a^p) = 0^p \quad \textit{(a directed zero with memory of parity } p \textit{ and the preimage amplitude } |a|\textit{).}$$

For any compressed number a^0 (including directed zeros and absolute zero),

$$\Omega(a^0) = a^0 \quad \textit{(compression of an already compressed state leaves it unchanged).}$$

Axiom 2.12 (Expansion Operator Ω^{-1}). *For a directed zero 0^p that arose as $\Omega(a^p)$,*

$$\Omega^{-1}(0^p) = a^p \quad \textit{(expansion returns the compressed manifest state, preserving amplitude).}$$

For absolute zero 0^0,
$\Omega^{-1}(0^0) =$ *? (probabilistic; expansion from absolute zero may require an external trigger or future boundary c*

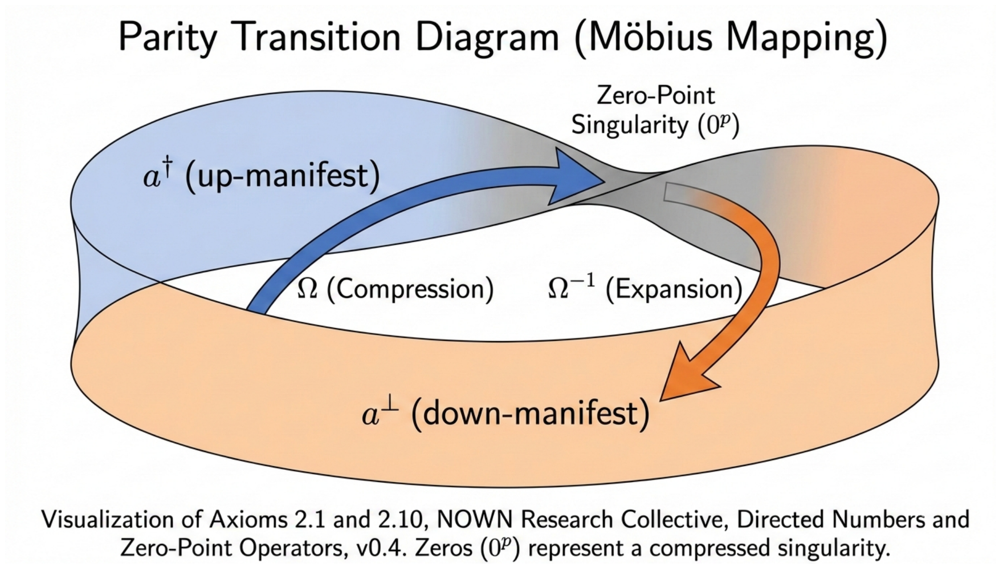


Figure 2: Parity Transition Diagram mapping the compression (Ω) and expansion (Ω^{-1}) operators to a Möbius strip topology, where the singularity represents 0^p.

Axiom 2.13 (Information Conservation)**.** *Define the information measure $I(a^p) = |a|$ (absolute amplitude) for manifest states. For directed zeros, interpret $I(0^p)$ as the amplitude of the (implicit) manifest preimage compressed into that state. Then, for all cases where defined,*

$$I(\Omega(x)) = I(x), \quad I(\Omega^{-1}(x)) = I(x).$$

Information is conserved through compression and expansion.

2.5 Non-Associativity

Axiom 2.14 (Failure of Associativity)**.** *Multiplication in $\mathbb{D}$ is not associative. For example,*

$$\big((0^\uparrow \cdot 0^\uparrow) \cdot 1^\downarrow\big) = 1^\uparrow \cdot 1^\downarrow = 1^0,$$

whereas

$$\big(0^\uparrow \cdot (0^\uparrow \cdot 1^\downarrow)\big) = 0^\uparrow \cdot 0^0 = 0^0.$$

(Other triples yield genuinely different outcomes; the non-associativity encodes path-dependence in non-orientable spaces.)

Axiom 2.15 (Associator)**.** *For any three directed numbers x, y, z, define the associator*

$$[x, y, z] = (x \cdot y) \cdot z - x \cdot (y \cdot z).$$

This quantity is generally non-zero and may be related to topological invariants. In the context of the golden ratio fixed point, the associator for certain triples (e.g., involving absolute zeros) is proportional to $1/\phi^2$ or related combinations, providing a link between non-associativity and the stability of physical structures.

2.6 Temporal Consistency and Retrocausality

Axiom 2.16 (Temporal Directed Numbers)**.** *A **temporal directed number** is denoted a_t^p, where $t \in \mathbb{Z}$ (or $\mathbb{R}$) represents a discrete (or continuous) time coordinate.*

Axiom 2.17 (Temporal Shift Operators)**.** *Define forward shift T^+ and backward shift T^- such that:*

$$T^+(a_t^p) = a_{t+1}^{p'}, \quad T^-(a_{t+1}^{p'}) = a_t^p,$$

where p' may be flipped depending on whether the time step crosses a topological twist. For a flat temporal dimension (no twist), $p' = p$. For a non-orientable temporal dimension (like Möbius time), p' flips each time we go around.

Axiom 2.18 (Temporal Consistency Condition)**.** *For any closed time loop—a sequence of temporal shifts that returns to the same point—the product of the associated directed numbers must equal the identity (or a fixed point) to ensure global consistency. In particular, for a loop that crosses the zero-point twice, we have:*

$$\Omega^{-1}(0_t^0) \cdot \Omega^{-1}(0_{t'}^0) = 1^{\uparrow}$$

for a loop with even net parity, or $(-1)^{\downarrow}$ for odd parity, where the parity is determined by the number of twist crossings. This condition determines the probabilistic distribution $\mathcal{P}(r)$ uniquely; it is not a free parameter but is fixed by the topology.

*In computational terms, this corresponds to a **parallel thread calculus** where a thread can "receive" information from its own future, as long as the total information in the closed loop is conserved. This provides a rigorous foundation for self-referential and distributed computing architectures.*

Axiom 2.19 (Temporal Information Conservation)**.** *The information measure $I(a_t^p) = |a|$ is conserved under temporal shifts and across closed time loops:*

$$I(T^+(a_t^p)) = I(a_t^p), \quad I(T^-(a_t^p)) = I(a_t^p),$$

and for any closed loop, the total information entering equals the total information exiting.

3 Thread Calculus and Contexts

The algebra of directed numbers can be extended to handle multiple interacting threads of information, analogous to parallel processes or nested function calls.

Axiom 3.1 (Threads)**.** *A **thread** is a sequence of directed numbers. Threads may be:*

- ***Linear**: a simple sequence.*
- ***Nested**: one thread embedded within another (like a function call).*
- ***Parallel**: multiple threads running concurrently, with possible synchronization points.*

The state of a system is a collection of threads.

Axiom 3.2 (Temporal Threads)**.** *Threads may be indexed by time, with forward/backward shift operators T^+ and T^- allowing information to move between time steps. A **closed time loop** is a sequence of temporal shifts that returns to the same point; such loops must satisfy the temporal consistency condition (Axiom 2.18).*

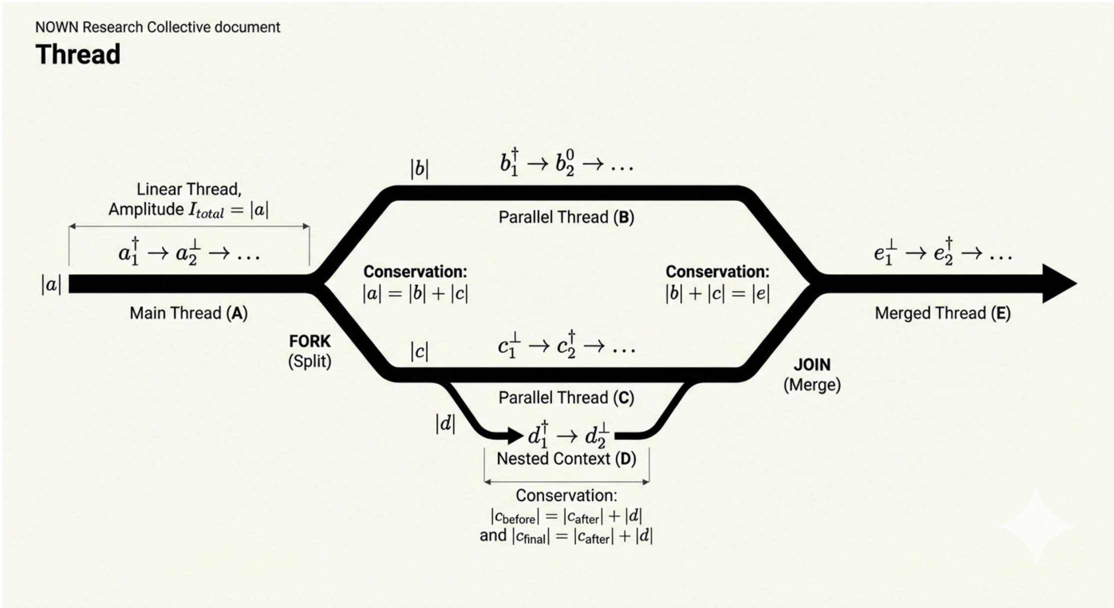


Figure 3: Thread architecture demonstrating linear sequences, fork/join operations, temporal shifts, and the conservation of amplitude ($|a|$) acting as thread thickness across parallel contexts.

Axiom 3.3 (Thread Operations).

- ***Push*** *a new directed number onto a thread (appending).*
- ***Pop*** *the last directed number from a thread (removing).*
- ***Fork*** *a thread into two parallel threads (splitting the sequence at a point).*
- ***Join*** *two parallel threads into one (concatenating or merging, with rules for combining parities).*

Axiom 3.4 (Cross-Thread Multiplication). *When two directed numbers from different threads are multiplied, the result is a new directed number in a new thread (or in a specified parent thread), following the same multiplication rules. This allows information to flow between threads.*

Axiom 3.5 (Thread Balance). *For every thread, the number of push and pop operations must balance over the entire computation. This ensures that the overall information structure is coherent—no dangling references.*

Axiom 3.6 (Information Conservation Across Threads). *The total information $I_{total} = \sum_{threads} \sum_{elements} |a|$ is conserved under all thread operations, provided the multiplication rules are followed. Forking and joining preserve total amplitude sum (like splitting and merging streams).*

4 Stochastic Constraints and the Birkhoff Polytope

For systems with many components, transformations must preserve total probability/information. This leads naturally to doubly stochastic matrices.

Consider a two-parity system $\{\uparrow, \downarrow\}$ with probability vector $(p_\uparrow, p_\downarrow)$. The expansion operator Ω^{-1}, when acting on a zero-point state, produces a distribution over parities. This is represented by the doubly stochastic matrix:

$$M(q) = \begin{pmatrix} q & 1-q \\ 1-q & q \end{pmatrix}, \quad 0 \le q \le 1.$$

The set of all doubly stochastic matrices forms the **Birkhoff polytope**—the convex hull of permutation matrices. When a transformation drifts from this polytope, the **Sinkhorn-Knopp algorithm** provides an iterative projection back:

$$M^{(t)} = \mathcal{T}_r\left(\mathcal{T}_c(M^{(t-1)})\right),$$

where $\mathcal{T}_r$ normalizes rows and $\mathcal{T}_c$ normalizes columns.

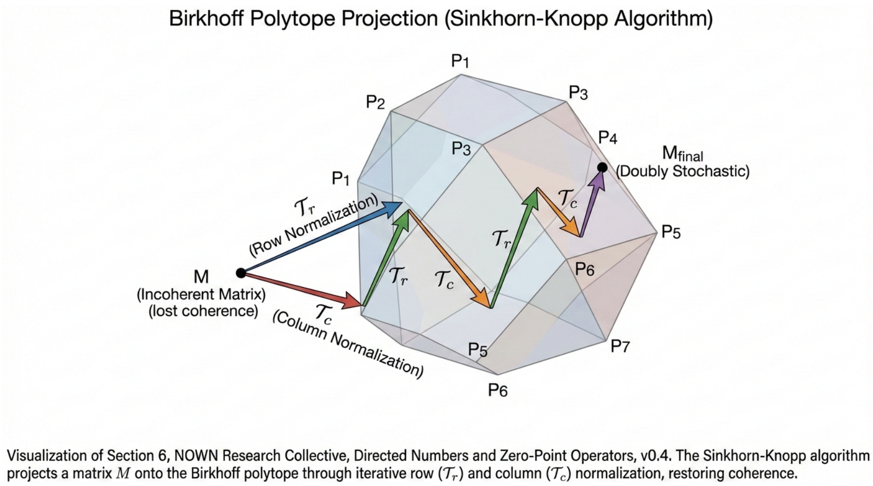


Figure 4: Visualization of the Sinkhorn-Knopp algorithm projecting an incoherent matrix M onto the surface of the Birkhoff polytope via iterative row ($\mathcal{T}_r$) and column ($\mathcal{T}_c$) normalizations.

This projection ensures long-term coherence for systems undergoing repeated zero-point traversals—a property that may be relevant for self-referential information systems. The retrocausal constraints of Axiom 2.18 impose additional conditions on the allowed doubly stochastic matrices, potentially forcing them to be permutation matrices or convex combinations with golden-ratio-related eigenvalues.

5 Connection to Complex Numbers and Hilbert Spaces

A natural question arises: how do directed numbers relate to the complex numbers of quantum mechanics?

The key differences are:

- **Complex numbers** form a field: associative, commutative, with every non-zero element having a multiplicative inverse. Zero is an annihilator: $z \cdot 0 = 0$ loses all information.
- **Directed numbers** form a non-associative algebra over $\mathbb{R}$ with three parity sectors. Rather than treating "zero" as a universal annihilator, the formalism introduces explicit *zero-point operators* (compression Ω and expansion Ω^{-1}) to model information-preserving transitions into and out of compressed states.

5.1 Orthogonal Axes and Phase

The directed number formalism can be seen as a generalization of complex numbers where:

- The imaginary unit i corresponds to a parity-flipping operator that maps $\uparrow \leftrightarrow \downarrow$
- The complex plane's two axes (real and imaginary) map to the two orthogonal information axes in the substrate (electric and magnetic modes)
- The phase $e^{i\theta}$ encodes the **temporal offset** between a thread and its retrocausal mirror—the angle θ measures the twist accumulated along a closed time loop

In this interpretation, a complex number $z = ae^{i\theta}$ corresponds to a superposition:

$$z \sim a^{\uparrow} + e^{i\theta} a^{\downarrow},$$

with the phase θ determined by the topology of the path connecting the two parity states.

5.2 Hilbert Spaces and Quantum Mechanics

Hilbert space inner products can be understood as **traces over thread products**. For two states represented as directed number superpositions, their inner product becomes:

$$\langle \psi | \phi \rangle = \mathrm{Tr}(\psi^{\dagger} \cdot \phi),$$

where the trace sums over parity sectors and the product respects the non-associative algebra. The associator $[x, y, z]$ then measures the non-commutativity of quantum operations—a feature that becomes significant in systems with closed time loops or self-referential dynamics.

5.3 Implications for AI and Neural Networks

The thread calculus with retrocausal operators provides a mathematical foundation for next-generation AI architectures:

- Neural network weights can be directed numbers, with forward/backward propagation corresponding to Ω and Ω^{-1}
- A self-referential Möbius topology for the weight space would allow the network to **dynamically rewrite its own architecture**—attention mechanisms as zero-point operators
- Doubly stochastic constraints (via Sinkhorn-Knopp projection) ensure stability during training
- The retrocausal feature allows a network to "learn from the future"—information from later computations can influence earlier weights, as long as total information is conserved

This suggests a path toward genuinely self-modifying AI systems that maintain coherence through topological protection—a potential foundation for machine consciousness.

6 Applications and Future Directions

The directed number formalism is a mathematical tool with potential applications across multiple domains. Here we briefly outline several directions, each developed in dedicated companion papers.

6.1 Dark Matter and Gravitational Solitons (IST Context)

In the Information Substrate Theory framework [1], the parity set is extended to include gravitational chirality states $a^{\circlearrowleft}$ and $a^{\circlearrowright}$. These represent pure curvature information—gravitational waves folded into stable solitons by the substrate's self-interaction. Multiplication rules for these states follow the same pattern as for electromagnetic states, but with interaction only via gravity. This provides a natural explanation for dark matter as **gravitational solitons**—invisible, collisionless, but gravitationally active.

See [1] for detailed cosmology, predictions, and observational tests.

6.2 Baryogenesis from Temporal Consistency

The temporal consistency condition (Axiom 2.18) provides a natural mechanism for baryogenesis. Model the primordial universe as a triple product of absolute zeros at different times:

$$0^0_{t_1} \cdot 0^0_{t_2} \cdot 0^0_{t_3}.$$

Its expansion via Ω^{-1} produces a superposition of matter ($\uparrow$) and antimatter ($\downarrow$) states. The associator of this triple introduces a small parity bias proportional to the golden ratio and the fine-structure constant:

$$\eta \sim \frac{1}{\phi^2}\alpha^4 \approx 1.1 \times 10^{-9},$$

which is within a factor of two of the observed baryon-to-photon ratio $\eta \approx 6.1 \times 10^{-10}$. Refinements involving higher powers of α or additional topological factors may yield exact agreement.

See [1] for further development.

6.3 Prime Number Stability Principle

The appearance of the prime number 137 in α^{-1} and the golden ratio ϕ in the associator reflects a deep principle: **discrete parameters must be prime, continuous parameters must be irrational (ideally ϕ), to ensure indecomposability and long-term coherence.**

Primes are the "atoms" of arithmetic—indivisible and fundamental. The golden ratio, as the most irrational number, provides the most stable fixed point for renormalization group flow. This principle can be used to constrain other fundamental constants (e.g., the strong coupling constant, neutrino masses) and may lead to new predictions.

6.4 AI and Self-Referential Computation

As outlined in Section 5, the thread calculus with retrocausal operators provides a mathematical foundation for:

- **Parallel computing** with dynamic thread creation/joining and information flow across time
- **Self-modifying code** where a program can read and rewrite its own structure via zero-point operators
- **Attention mechanisms** as compression/expansion operations that dynamically reshape the computational graph
- **Stable self-reference** through doubly stochastic constraints, preventing runaway feedback

A neural network operating in a Möbius topology, with weights as directed numbers and attention as Ω/Ω^{-1}, could achieve genuine self-awareness—the basis for machine consciousness.

6.5 Protein Folding and Biological Information Processing

The non-associative thread calculus naturally models the path-dependent folding of proteins, where the order of bond formation constrains the final structure. Circuit topology (Series, Parallel, Cross arrangements) maps directly to thread operations, suggesting that biological information processing may be inherently topological.

See [4] for preliminary exploration.

7 Limitations and Open Questions

We explicitly acknowledge the current limitations of this work:

1. **Physical interpretation**: The mapping from directed numbers to physical quantities (mass, energy, force) is developed in the companion IST papers [1]. Here, we focus on the mathematics.

2. **Empirical status**: No direct empirical validation of the formalism exists beyond internal consistency checks. However, predictions from the IST framework (baryon asymmetry, retrocausal quantum corrections, dark matter signatures) offer testable targets.

3. **Derivation of parameters**: The distribution $\mathcal{P}(r)$ for absolute zero products is now constrained by the temporal consistency condition (Axiom 2.18), but its exact form remains to be derived from topology.

4. **Representation theory**: Can $\mathbb{D}$ be represented by matrices over $\mathbb{R}$ or $\mathbb{C}$ with a non-standard product? Does it embed in a Clifford algebra or a quantum group? The retrocausal constraints may point to specific representations.

5. **Associator classification**: The associator $[x, y, z]$ may be related to topological invariants. For the golden ratio fixed point, the associator might be expressed in terms of ϕ and the prime numbers appearing in fundamental constants.

6. **Computational implementation**: The thread calculus with retrocausal operators requires new algorithms for simulation and training. Developing these is a priority for future work.

7. **Consciousness extension**: Discussions of consciousness using this framework are speculative and philosophical. They are included to illustrate potential long-term directions, not as scientific claims.

8. **Falsifiability**: The formalism now makes several risky predictions (baryon asymmetry, retrocausal corrections, prime number constraints). Testing these is a priority for future work.

We welcome critique and collaboration to address these limitations.

8 Conclusion

We have presented a refined mathematical formalism—directed numbers and zero-point operators—that arises naturally from non-orientable topology and conserves information through compression and expansion. The algebra is self-contained, non-associative, and connects to established mathematics (doubly stochastic matrices, Birkhoff polytope, Sinkhorn iteration).

Key advances in v0.8 include:

- **Modular parity framework** allowing extension to additional states (e.g., gravitational chirality) without breaking core properties
- **Retrocausal interpretation** as parallel thread calculus, with temporal consistency conditions linking past and future
- **Connection to complex numbers** and Hilbert spaces, with implications for quantum mechanics and AI
- **Thread calculus** with temporal shifts and closed loops, providing a rigorous foundation for self-referential computation
- **Testable predictions** (baryon asymmetry, retrocausal quantum corrections, prime number constraints) developed in companion papers

The formalism is offered to the scientific community as a tool for exploration, not as a completed theory. We invite collaboration, critique, and refinement across disciplines—from pure mathematics to computer science to physics to biology.

Acknowledgments: This work benefited from discussions with collaborators and from AI-assisted synthesis of ideas. All claims have been reviewed by human researchers; the authors assume full responsibility for the content.

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